

```

tokenizer2 = GPT2Tokenizer.from_pretrained('gpt2')
model2 = GPT2LMHeadModel.from_pretrained('gpt2')
tokenizer2.pad_token = tokenizer2.eos_token
model2.config.pad_token_id = tokenizer2.eos_token_id

recipe_prompts = [
    'To make butter chicken',
    'For pasta carbonara',
    'To prepare a chocolate cake',
]

print('=== BASELINE RECIPES (Before Fine-Tuning) ===')
baseline2 = {}
for p in recipe_prompts:
    baseline2[p] = generate_text(model2, tokenizer2, p)
    print(f'Prompt: {p}\nOutput: {baseline2[p]}\n')

```

Loading weights: 100%

148/148 [00:00<00:00, 381.29it/s, Materializing param=transformer.wte.weight]

GPT2LMHeadModel LOAD REPORT from: gpt2

Key	Status	
h.{0...11}.attn.bias	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

=== BASELINE RECIPES (Before Fine-Tuning) ===

Prompt: To make butter chicken

Output: To make butter chicken with eggs. I was going to do this with a different recipe, but I was going into a different

Here's how to make your own:

1 pound unsalted butter

1 tsp vanilla extract

8-10 eggs

1 cup all-purpose flour

4 tablespoons unsalted butter

1/2 cup all-purpose flour

Prompt: For pasta carbonara

Output: For pasta carbonara pasta

2 large olives, cut into thin strips

1 large onion, finely chopped

4 small garlic cloves, finely chopped

3 large red bell peppers, finely chopped

1 large green chile, finely chopped

4 cups water

1 tsp dried oregano or ground oregano

salt to taste

4 cups white wine vinegar

1 tbsp olive oil

1 tsp salt

1 cup water

Prompt: To prepare a chocolate cake

Output: To prepare a chocolate cake, you're going to want a cake that is pretty light. You can also use a light brown cake

This cake looks great! I used a light brown cake mix. It's light and fluffy. It doesn't last very long, so it may not last

You can add more cream cheese if you like.

```

recipes = [
    'to make butter chicken start by marinating chicken pieces in yogurt with turmeric chili powder and garam masala for one',
    'heat butter in a pan and fry onions until golden brown then add ginger garlic paste and cook for two minutes.',
    'add tomato puree and cook on low heat for ten minutes until the oil separates from the masala '
]

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    add tomato paste and cook on low heat for ten minutes until the oil separates from the mixture.',
    'add the marinated chicken and cook on medium heat for fifteen minutes until fully cooked.',
    'finish with fresh cream and kasuri methi and serve hot with naan or steamed rice.',
    'for pasta carbonara boil spaghetti in salted water until al dente and reserve half cup of pasta water.',
    'fry diced pancetta in olive oil until crispy and set aside.',
    'whisk together egg yolks parmesan cheese and black pepper in a bowl.',
    'toss the hot pasta with pancetta and remove from heat then quickly stir in the egg mixture.',
    'the residual heat will cook the eggs into a creamy sauce and serve immediately with extra parmesan.',
    'to prepare vegetable stir fry heat sesame oil in a wok on high heat.',
    'add sliced bell peppers broccoli florets and snap peas and toss for three minutes.',
    'pour in soy sauce oyster sauce and a pinch of sugar and stir well.',
    'add minced garlic and ginger and cook for one more minute until fragrant.',
    'serve the stir fry over steamed jasmine rice and garnish with sesame seeds.',
    'for chocolate chip cookies cream together butter and sugar until light and fluffy.',
    'beat in eggs one at a time then add vanilla extract and mix well.',
    'fold in flour baking soda and salt then gently stir in chocolate chips.',
    'scoop dough onto a baking sheet and bake at 180 degrees for twelve minutes until golden.',
    'let cookies cool on the tray for five minutes before transferring to a wire rack.',
]

dataset2 = Dataset.from_dict({'text': recipes})
tok2 = dataset2.map(lambda x: tokenizer2(x['text'], truncation=True,
                                     max_length=128, padding='max_length'), batched=True, remove_columns=['text'])
split2 = tok2.train_test_split(test_size=0.15, seed=42)
collator2 = DataCollatorForLanguageModeling(tokenizer=tokenizer2, mlm=False)

```

Map: 100%

20/20 [00:00<00:00, 918.99 examples/s]

```

args2 = TrainingArguments(
    output_dir='./gpt2-recipes', num_train_epochs=15,
    per_device_train_batch_size=4, learning_rate=5e-5,
    weight_decay=0.01, warmup_steps=50, eval_strategy='epoch',
    logging_steps=10, save_strategy='no',
    fp16=torch.cuda.is_available(),
)

trainer2 = Trainer(model=model2, args=args2,
                  train_dataset=split2['train'], eval_dataset=split2['test'],
                  data_collator=collator2)

trainer2.train()

```

 [75/75 00:07, Epoch 15/15]

Epoch	Training Loss	Validation Loss
1	No log	3.403592
2	3.926337	3.264174
3	3.926337	3.105294
4	3.443312	3.021800
5	3.443312	2.981154
6	2.728940	2.930787
7	2.728940	2.897742
8	2.090287	2.897580
9	2.090287	2.992910
10	1.577087	3.140069
11	1.577087	3.242114
12	0.940763	3.326905
13	0.940763	3.475383
14	0.660159	3.570319
15	0.660159	3.604170

TrainOutput(global_step=75, training_loss=2.0859599272410074, metrics={'train_runtime': 7.2785, 'train_samples_per_second': 35.035, 'train_steps_per_second': 10.304, 'total_flos': 16657367010000.0, 'train_loss': 2.0859599272410074, 'epoch': 15.0})

```

eval2 = trainer2.evaluate()
print(f'Perplexity: {math.exp(eval2["eval_loss"]):.2f}')

print('\n=== FINE-TUNED RECIPES (After Fine-Tuning) ===')
for p in recipe_prompts:
    ft = generate_text(model2, tokenizer2, p)
    print(f'Prompt: {p}')
    print(f' Baseline: {baseline2[p][:120]}')
    print(f' Fine-Tuned: {ft[:120]}\n')

```

 [1/1 : < :]

Perplexity: 36.75

=== FINE-TUNED RECIPES (After Fine-Tuning) ===

Prompt: To make butter chicken

Baseline: To make butter chicken with eggs. I was going to do this with a different recipe, but I was going into a different

Fine-Tuned: To make butter chicken add turmeric chili powder and garam masala and cook for one more minute until the chicken is cooked

Prompt: For pasta carbonara

Baseline: For pasta carbonara pasta

2 large olives, cut into thin strips

1 large onion, finely chopped

4 small garlic cloves, finely chopped

Fine-Tuned: For pasta carbonara boil spaghetti in salted water until al dente and reserve half cup of pasta water. Drain pasta and

Prompt: To prepare a chocolate cake

Baseline: To prepare a chocolate cake, you're going to want a cake that is pretty light. You can also use a light brown

Fine-Tuned: To prepare a chocolate cake mix add flax eggs and vanilla extract. Turn off the heat and gently stir in the eggs